



Reg. No:

|  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|

**SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE**

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

**AUTONOMOUS EXAMINATIONS - MAY'2025**

**COMMON TO B.E – CSE & M.TECH - CSE**

**20CS2E11 – BIG DATA TECHNOLOGIES**

**Answer ALL questions**

**Duration: 2 Hrs 30 min**

**Maximum: 70 Marks**

**PART - A**

**(5x 1 = 5 Marks)**

1. **CO1 The data from the CCTV Coverage and weather forecast report is of \_\_\_\_\_.**  
a) Structured Data                      b) Semi-structured Data  
c) Unstructured Data                      d) Structured and Semi-structured Data
2. **CO1 The typical database server scales \_\_\_\_\_.**  
a) Distributed File System                      b) Central Server  
c) Single File system                      d) Two or more systems
3. **CO2 Which of these provides a Stream processing system used in Hadoop ecosystem?**  
a) Hive                      b) Solr                      c) Spark                      d) Tez
4. **CO3 Hadoop is a framework that works with a variety of related tools. Common cohorts include \_\_\_\_\_.**  
a) MapReduce, Hive and HBase                      b) MapReduce, MySQL and Google Apps  
c) MapReduce, Hummer and Iguana                      d) MapReduce, Heron and Trumpe
5. **CO3 A \_\_\_\_\_ node acts as the Slave and is responsible for executing a Task assigned to it by the Job Tracker.**  
a) Mapreducer                      b) Mapper                      c) Tasktracker                      d) Jobtracker

**PART - B**

**(5x 3 = 15 Marks)**

6. **CO1** Discuss big data in healthcare and medicine.
7. **CO3** Why Hadoop is called a Big Data technology?
8. **CO4** List the types of NoSQL Databases and explain any one.
9. **CO3** What is compaction in HBase?
10. **CO2** Explain about the load and store functions in Pig.

**PART - C**

**(5 x 10 = 50 Marks)**

**Q.No: 11 Compulsory Question**

**Q.No: 12 to 17 Answer any FOUR Questions**

**Compulsory Question:**

11. **CO2** i) Explain HDFS Architecture with a neat diagram. (5)  
**CO2** ii) Difference between RDBMS and Big Data. (5)
12. **CO1** i) What are the 5 Vs of Big Data? (5)  
**CO1** ii) List out the challenges of big data. (5)
13. **CO3** i) How does shuffling and sorting work in MapReduce? (5)  
**CO3** ii) Difference between MapReduce 1 and MapReduce 2 (YARN). (5)
14. **CO4** i) How does CAP Theorem apply to NoSQL databases? (5)  
**CO4** ii) Elaborate the types of NOSQL briefly. (5)
15. **CO5** i) Explain the architecture of Zookeeper. (5)  
**CO5** ii) State the advantages of NoSQL. (5)
16. **CO5** i) What are the core components of Flume? (5)  
**CO3** ii) What is the role of the JobTracker and TaskTracker in Hadoop 1.x? (5)
17. **CO1** i) Discuss about CAP theorem with a suitable example. (5)  
**CO2** ii) Explain the modes in Pig runs. (5)

\*\*\*\*\*